

Three Friction Layers Slowing AI Proliferation

Physical Layer

Resolution: 5-10 years

Power

134GW needed by 2030 · 7-yr grid queue · 50% delayed

Chips

TSMC capacity = 1/3 demand · New fabs 3-5 yrs · \$20B+

Robot Hardware

Battery 2-3h · No safety standards · Dexterity gaps

Training Data

High-quality text exhausted ~2028 · Synthetic data partial fix

Carbon Emissions

AI: 32.8-79.7M tons CO2/yr · Cities issuing bans

Trust Layer

Resolution: 2-5 years

Hallucination

Factual QA hallucination 33-48% · Legal false citations rampant

Last Mile Problem

80% to 99.9% takes 80% of effort · Waymo still geo-fenced

ROI Trough

95% enterprise GenAI zero ROI · Avg burn \$1.9M/project

Historical Precedent

Self-driving: 2015 promise, 2026 still geo-fenced · GM Cruise burned \$10B then shut down · Fusion: always "30 yrs away" · Metaverse: 38 daily users

Institutional Layer

Resolution: 3-7+ years

Regulation

EU AI Act fully effective 2026 · China cybersecurity law covers AI

Unions

Ports ban full automation · WGA/SAG limit AI · Dock strikes

Liability Gap

AI agents have no legal personhood · No federal robot safety rules

Software scales in milliseconds. Power plants take 5 years. White-collar winter arrives before blue-collar.

These frictions slow the pace, not the direction. Every bottleneck has a clear resolution path.